Cumulant expansion of the three-body terms

generated by `cumulant_expansion.py`

The three-body operators surviving the space restriction are decomposed into two-body and one-body operators (plus scalar terms) with density-matrix coefficients, using the spin-free cumulant decomposition of `cumulant_decomposition/spin_free/spin_free.tex`:

$$\begin{aligned}
\hat{E}^{pqr}_{stu} \approx = {} & D^p_s \hat{E}^{qr}_{tu} - \tfrac{1}{2}\, D^p_t \hat{E}^{qr}_{su} - \tfrac{1}{2}\, D^p_u \hat{E}^{qr}_{ts} && (1)\\
& - \tfrac{1}{2}\, D^q_s \hat{E}^{pr}_{tu} + D^q_t \hat{E}^{pr}_{su} - \tfrac{1}{2}\, D^q_u \hat{E}^{pr}_{st} && (2)\\
& - \tfrac{1}{2}\, D^r_s \hat{E}^{qp}_{tu} - \tfrac{1}{2}\, D^r_t \hat{E}^{pq}_{su} + D^r_u \hat{E}^{pq}_{st} + D^{pq}_{st} \hat{E}^r_u && (3)\\
& - \tfrac{1}{2}\, D^{pq}_{su} \hat{E}^r_t - \tfrac{1}{2}\, D^{pq}_{ut} \hat{E}^r_s - \tfrac{1}{2}\, D^{pr}_{st} \hat{E}^q_u && (4)\\
& + D^{pr}_{su} \hat{E}^q_t - \tfrac{1}{2}\, D^{pr}_{tu} \hat{E}^q_s - \tfrac{1}{2}\, D^{qr}_{su} \hat{E}^p_t + D^{qr}_{tu} \hat{E}^p_s && (5)\\
& - \tfrac{1}{2}\, D^{rq}_{st} \hat{E}^p_u - 2\, D^p_s D^q_t \hat{E}^r_u - 2\, D^p_s D^r_u \hat{E}^q_t && (6)\\
& - \tfrac{1}{2}\, D^p_t D^q_u \hat{E}^r_s - \tfrac{1}{2}\, D^p_t D^r_s \hat{E}^q_u - \tfrac{1}{2}\, D^p_u D^q_s \hat{E}^r_t && (7)\\
& - \tfrac{1}{2}\, D^p_u D^r_t \hat{E}^q_s - \tfrac{1}{2}\, D^q_s D^r_t \hat{E}^p_u - 2\, D^q_t D^r_u \hat{E}^p_s && (8)\\
& - \tfrac{1}{2}\, D^q_u D^r_s \hat{E}^p_t + D^p_s D^q_u \hat{E}^r_t + D^p_s D^r_t \hat{E}^q_u && (9)\\
& + D^p_t D^q_s \hat{E}^r_u + D^p_t D^r_u \hat{E}^q_s + D^p_u D^q_t \hat{E}^r_s + D^p_u D^r_s \hat{E}^q_t && (10)\\
& + D^q_s D^r_u \hat{E}^p_t + D^q_t D^r_s \hat{E}^p_u + D^q_u D^r_t \hat{E}^p_s - D^p_s D^{qr}_{tu} && (11)\\
& + \tfrac{1}{2}\, D^p_t D^{qr}_{su} + \tfrac{1}{2}\, D^p_u D^{rq}_{st} + \tfrac{1}{2}\, D^q_s D^{pr}_{tu} - D^q_t D^{pr}_{su} && (12)\\
& + \tfrac{1}{2}\, D^q_u D^{pr}_{st} + \tfrac{1}{2}\, D^r_s D^{pq}_{ut} + \tfrac{1}{2}\, D^r_t D^{pq}_{su} - D^r_u D^{pq}_{st} && (13)\\
& + 4\, D^p_s D^q_t D^r_u - 2\, D^p_s D^q_u D^r_t - 2\, D^p_t D^q_s D^r_u + D^p_t D^q_u D^r_s && (14)\\
& + D^p_u D^q_s D^r_t - 2\, D^p_u D^q_t D^r_s && (15)
\end{aligned}$$

This is an operator identity: the labels p, q, r, s, t, u are fixed on both sides, so no index may be renamed; only the index pairs of each operator can be permuted together (the diagonal pair-permutation symmetry of the spin-summed $\hat{E}$, no sign). With the terms grouped by the operator in this sense, the decomposition reads

$$\begin{aligned}
\hat{E}^{pqr}_{stu} \approx {} & \hat{E}^{pq}_{st}\ (D^r_u) && (16)\\
& \hat{E}^{pq}_{su}\ \left(-\tfrac{1}{2}\, D^r_t\right) && (17)\\
& \hat{E}^{pq}_{ut}\ \left(-\tfrac{1}{2}\, D^r_s\right) && (18)\\
& \hat{E}^{pr}_{st}\ \left(-\tfrac{1}{2}\, D^q_u\right) && (19)\\
& \hat{E}^{pr}_{su}\ (D^q_t) && (20)\\
& \hat{E}^{pr}_{tu}\ \left(-\tfrac{1}{2}\, D^q_s\right) && (21)\\
& \hat{E}^{qr}_{su}\ \left(-\tfrac{1}{2}\, D^p_t\right) && (22)
\end{aligned}$$

$$\hat{E}^{qr}_{ts}\ \left(-\tfrac{1}{2}\,D^{p}_{u}\right) \tag{23}$$
$$\hat{E}^{qr}_{tu}\ \left(D^{p}_{s}\right) \tag{24}$$
$$\hat{E}^{p}_{s}\ \left(D^{qr}_{tu} - 2\,D^{q}_{t}D^{r}_{u} + D^{q}_{u}D^{r}_{t}\right) \tag{25}$$
$$\hat{E}^{p}_{t}\ \left(-\tfrac{1}{2}\,D^{qr}_{su} + D^{q}_{s}D^{r}_{u} - \tfrac{1}{2}\,D^{q}_{u}D^{r}_{s}\right) \tag{26}$$
$$\hat{E}^{p}_{u}\ \left(-\tfrac{1}{2}\,D^{q}_{s}D^{r}_{t} + D^{q}_{t}D^{r}_{s} - \tfrac{1}{2}\,D^{rq}_{st}\right) \tag{27}$$
$$\hat{E}^{q}_{s}\ \left(-\tfrac{1}{2}\,D^{pr}_{tu} + D^{p}_{t}D^{r}_{u} - \tfrac{1}{2}\,D^{p}_{u}D^{r}_{t}\right) \tag{28}$$
$$\hat{E}^{q}_{t}\ \left(D^{pr}_{su} - 2\,D^{p}_{s}D^{r}_{u} + D^{p}_{u}D^{r}_{s}\right) \tag{29}$$
$$\hat{E}^{q}_{u}\ \left(-\tfrac{1}{2}\,D^{pr}_{st} + D^{p}_{s}D^{r}_{t} - \tfrac{1}{2}\,D^{p}_{t}D^{r}_{s}\right) \tag{30}$$
$$\hat{E}^{r}_{s}\ \left(-\tfrac{1}{2}\,D^{pq}_{ut} - \tfrac{1}{2}\,D^{p}_{t}D^{q}_{u} + D^{p}_{u}D^{q}_{t}\right) \tag{31}$$
$$\hat{E}^{r}_{t}\ \left(-\tfrac{1}{2}\,D^{pq}_{su} + D^{p}_{s}D^{q}_{u} - \tfrac{1}{2}\,D^{p}_{u}D^{q}_{s}\right) \tag{32}$$
$$\hat{E}^{r}_{u}\ \left(D^{pq}_{st} - 2\,D^{p}_{s}D^{q}_{t} + D^{p}_{t}D^{q}_{s}\right) \tag{33}$$
$$- D^{p}_{s}D^{qr}_{tu} + 4\,D^{p}_{s}D^{q}_{t}D^{r}_{u} - 2\,D^{p}_{s}D^{q}_{u}D^{r}_{t} + \tfrac{1}{2}\,D^{p}_{t}D^{qr}_{su} \tag{34}$$
$$- 2\,D^{p}_{t}D^{q}_{s}D^{r}_{u} + D^{p}_{t}D^{q}_{u}D^{r}_{s} + D^{p}_{u}D^{q}_{s}D^{r}_{t} \tag{35}$$
$$- 2\,D^{p}_{u}D^{q}_{t}D^{r}_{s} + \tfrac{1}{2}\,D^{p}_{u}D^{rq}_{st} + \tfrac{1}{2}\,D^{q}_{s}D^{pr}_{tu} \tag{36}$$
$$- D^{q}_{t}D^{pr}_{su} + \tfrac{1}{2}\,D^{q}_{u}D^{pr}_{st} + \tfrac{1}{2}\,D^{r}_{s}D^{pq}_{ut} \tag{37}$$
$$+ \tfrac{1}{2}\,D^{r}_{t}D^{pq}_{su} - D^{r}_{u}D^{pq}_{st} \tag{38}$$

(19 operator groups; a group with no $\hat{E}$ shown is the scalar part.) The $\approx$ drops the normal-ordered three-body remainder and the three-cumulant λ^{pqr}_{stu}; for a single-determinant reference $\lambda_3 = 0$ identically. Every density matrix carrying a virtual index vanishes; density matrices with occupied or whole-space indices are kept symbolic as D^{i}_{j}, D^{ij}_{kl}, ... (no delta-function substitution). The formula is verified numerically for a closed-shell single-determinant reference on Jordan–Wigner matrices (random index assignments with and without the virtual rule).

1 Final blocks

Only the blocks containing surviving three-body terms change; the three-body pieces are decomposed and merged back into the block. Blocks not shown are unchanged from `space_restriction.tex`.

$\left[\hat{V},\,\hat{T}_2\right]$

after the space restriction (2 terms, 1 three-body):

$$\left[\hat{V},\,\hat{T}_2\right] = \tfrac{1}{2}\,g^{a_1a_2}_{p_1p_2}t^{a_1a_2}_{i_1i_2}\hat{E}^{p_1p_2}_{i_1i_2} \tag{39}$$
$$+\ g^{a_2p_3}_{p_1p_2}t^{a_2a_1}_{i_2i_1}\hat{E}^{a_1p_1p_2}_{i_1i_2p_3} \tag{40}$$

each three-body term is decomposed with the grouped identity. Every template index appears in exactly one factor of each piece, so a whole operator group survives or dies together: a group survives iff its operator carries every virtual index of the pattern; in particular the scalar part (pure density matrices) always vanishes. The surviving groups are: 9 of the 19 groups survive (6 two-body, 3 one-body):

$$g^{a_2p_3}_{p_1p_2}t^{a_2a_1}_{i_2i_1}\hat{E}^{a_1p_1p_2}_{i_1i_2p_3} \approx \hat{E}^{a1p1}_{i1i2}\ \left(g^{a_2p_3}_{p_1p_2}t^{a_2a_1}_{i_2i_1}D^{p_2}_{p_3}\right) \tag{41}$$

$$\hat{E}^{a1p1}_{i1p3}\ \left(-\tfrac{1}{2}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_2}_{i_2}\right) \tag{42}$$
$$\hat{E}^{a1p1}_{p3i2}\ \left(-\tfrac{1}{2}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_2}_{i_1}\right) \tag{43}$$
$$\hat{E}^{a1p2}_{i1i2}\ \left(-\tfrac{1}{2}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{p_3}\right) \tag{44}$$
$$\hat{E}^{a1p2}_{i1p3}\ \left(g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_2}\right) \tag{45}$$
$$\hat{E}^{a1p2}_{i2p3}\ \left(-\tfrac{1}{2}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_1}\right) \tag{46}$$
$$\hat{E}^{a1}_{i1}\ \Big(g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1p_2}_{i_2p_3} \tag{47}$$
$$-2\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_2} D^{p_2}_{p_3} \tag{48}$$
$$+g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{p_3} D^{p_2}_{i_2}\Big) \tag{49}$$
$$\hat{E}^{a1}_{i2}\ \Big(-\tfrac{1}{2}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1p_2}_{i_1p_3} \tag{50}$$
$$+g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_1} D^{p_2}_{p_3} \tag{51}$$
$$-\tfrac{1}{2}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{p_3} D^{p_2}_{i_1}\Big) \tag{52}$$
$$\hat{E}^{a1}_{p3}\ \left(-\tfrac{1}{2}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_1} D^{p_2}_{i_2}\right. \tag{53}$$
$$+g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_2} D^{p_2}_{i_1} \tag{54}$$
$$\left.-\tfrac{1}{2}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_2p_1}_{i_1i_2}\right) \tag{55}$$

the decomposition of the three-body part produces 15 terms; the final block (surviving terms and decomposition merged) has 16 terms:

$$\left[\hat{V},\, \hat{T}_2\right] = -\tfrac{1}{2}\, D^{i_2p_1}_{p_2p_3} g^{a_2p_1}_{p_2p_3} t^{a_2a_1}_{i_1i_2} \hat{E}^{a_1}_{i_1} \tag{56}$$
$$+\, D^{i_2p_1}_{p_2p_3} g^{a_2p_1}_{p_2p_3} t^{a_2a_1}_{i_2i_1} \hat{E}^{a_1}_{i_1} \tag{57}$$
$$+\, D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_2}_{p_1p_3} t^{a_2a_1}_{i_1i_2} \hat{E}^{a_1}_{i_1} \tag{58}$$
$$-\, 2\, D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_2}_{p_1p_3} t^{a_2a_1}_{i_2i_1} \hat{E}^{a_1}_{i_1} \tag{59}$$
$$-\, \tfrac{1}{2}\, D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_2}_{p_3p_1} t^{a_2a_1}_{i_1i_2} \hat{E}^{a_1}_{i_1} \tag{60}$$
$$+\, D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_2}_{p_3p_1} t^{a_2a_1}_{i_2i_1} \hat{E}^{a_1}_{i_1} \tag{61}$$
$$-\, \tfrac{1}{2}\, D^{i_1i_2}_{p_2p_3} g^{a_2p_1}_{p_2p_3} t^{a_2a_1}_{i_1i_2} \hat{E}^{a_1}_{p_1} \tag{62}$$
$$+\, D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2p_1}_{p_2p_3} t^{a_2a_1}_{i_1i_2} \hat{E}^{a_1}_{p_1} \tag{63}$$
$$-\, \tfrac{1}{2}\, D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2p_1}_{p_2p_3} t^{a_2a_1}_{i_2i_1} \hat{E}^{a_1}_{p_1} \tag{64}$$
$$+\, D^{p_2}_{p_3} g^{a_2p_2}_{p_1p_3} t^{a_2a_1}_{i_2i_1} \hat{E}^{a_1p_1}_{i_1i_2} \tag{65}$$
$$-\, \tfrac{1}{2}\, D^{p_2}_{p_3} g^{a_2p_2}_{p_3p_1} t^{a_2a_1}_{i_2i_1} \hat{E}^{a_1p_1}_{i_1i_2} \tag{66}$$
$$-\, \tfrac{1}{2}\, D^{i_2}_{p_3} g^{a_2p_2}_{p_1p_3} t^{a_2a_1}_{i_2i_1} \hat{E}^{a_1p_1}_{i_1p_2} \tag{67}$$
$$-\, \tfrac{1}{2}\, D^{i_2}_{p_3} g^{a_2p_2}_{p_3p_1} t^{a_2a_1}_{i_1i_2} \hat{E}^{a_1p_1}_{i_1p_2} \tag{68}$$
$$+\, D^{i_2}_{p_3} g^{a_2p_2}_{p_3p_1} t^{a_2a_1}_{i_2i_1} \hat{E}^{a_1p_1}_{i_1p_2} \tag{69}$$
$$-\, \tfrac{1}{2}\, D^{i_2}_{p_3} g^{a_2p_2}_{p_1p_3} t^{a_2a_1}_{i_1i_2} \hat{E}^{a_1p_1}_{p_2i_1} \tag{70}$$
$$+\, \tfrac{1}{2}\, g^{a_1a_2}_{p_1p_2} t^{a_1a_2}_{i_1i_2} \hat{E}^{p_1p_2}_{i_1i_2} \tag{71}$$

grouped by the operator, so that all coefficient pieces multiplying the same $\hat{E}$ can be substituted together (6 operator groups):

$$\left[\hat{V},\,\hat{T}_2\right] = \hat{E}^{a1}_{i1}\ \big(-\tfrac{1}{2}\,D^{i_2p_1}_{p_2p_3}g^{a_2p_1}_{p_2p_3}t^{a_2a_1}_{i_1i_2} \tag{72}$$

$$+D^{i_2p_1}_{p_2p_3}g^{a_2p_1}_{p_2p_3}t^{a_2a_1}_{i_2i_1} \tag{73}$$

$$+D^{i_2}_{p_1}D^{p_2}_{p_3}g^{a_2p_2}_{p_1p_3}t^{a_2a_1}_{i_1i_2} \tag{74}$$

$$-2\,D^{i_2}_{p_1}D^{p_2}_{p_3}g^{a_2p_2}_{p_1p_3}t^{a_2a_1}_{i_2i_1} \tag{75}$$

$$-\tfrac{1}{2}\,D^{i_2}_{p_1}D^{p_2}_{p_3}g^{a_2p_2}_{p_3p_1}t^{a_2a_1}_{i_1i_2} \tag{76}$$

$$+D^{i_2}_{p_1}D^{p_2}_{p_3}g^{a_2p_2}_{p_3p_1}t^{a_2a_1}_{i_2i_1}\big) \tag{77}$$

$$\hat{E}^{a1}_{p1}\ \big(-\tfrac{1}{2}\,D^{i_1i_2}_{p_2p_3}g^{a_2p_1}_{p_2p_3}t^{a_2a_1}_{i_1i_2} \tag{78}$$

$$+D^{i_1}_{p_2}D^{i_2}_{p_3}g^{a_2p_1}_{p_2p_3}t^{a_2a_1}_{i_1i_2} \tag{79}$$

$$-\tfrac{1}{2}\,D^{i_1}_{p_2}D^{i_2}_{p_3}g^{a_2p_1}_{p_2p_3}t^{a_2a_1}_{i_2i_1}\big) \tag{80}$$

$$\hat{E}^{p1a1}_{i1p2}\ \big(-\tfrac{1}{2}\,D^{i_2}_{p_3}g^{a_2p_2}_{p_1p_3}t^{a_2a_1}_{i_1i_2}\big) \tag{81}$$

$$\hat{E}^{p1a1}_{i2i1}\ \big(D^{p_2}_{p_3}g^{a_2p_2}_{p_1p_3}t^{a_2a_1}_{i_2i_1} \tag{82}$$

$$-\tfrac{1}{2}\,D^{p_2}_{p_3}g^{a_2p_2}_{p_3p_1}t^{a_2a_1}_{i_2i_1}\big) \tag{83}$$

$$\hat{E}^{p1a1}_{p2i1}\ \big(-\tfrac{1}{2}\,D^{i_2}_{p_3}g^{a_2p_2}_{p_1p_3}t^{a_2a_1}_{i_2i_1} \tag{84}$$

$$-\tfrac{1}{2}\,D^{i_2}_{p_3}g^{a_2p_2}_{p_3p_1}t^{a_2a_1}_{i_1i_2} \tag{85}$$

$$+D^{i_2}_{p_3}g^{a_2p_2}_{p_3p_1}t^{a_2a_1}_{i_2i_1}\big) \tag{86}$$

$$\hat{E}^{p1p2}_{i1i2}\ \big(\tfrac{1}{2}\,g^{a_1a_2}_{p_1p_2}t^{a_1a_2}_{i_1i_2}\big) \tag{87}$$

$$\left[\left[\hat{F},\,\hat{T}_2\right]^{\dagger},\,\hat{T}_2\right]$$

after the space restriction (6 terms, 4 three-body):

$$\left[\left[\hat{F},\,\hat{T}_2\right]^{\dagger},\,\hat{T}_2\right] = f^{a_1}_{a_2}t^{a_1a_3}_{i_1i_2}t^{a_2a_3}_{i_3i_4}\,\hat{E}^{i_1i_2}_{i_3i_4} \tag{88}$$

$$-\,f^{i_4}_{p_1}t^{a_1a_2}_{i_1i_4}t^{a_1a_2}_{i_2i_3}\,\hat{E}^{i_1p_1}_{i_2i_3} \tag{89}$$

$$+\,f^{a_3}_{a_4}t^{a_3a_1}_{i_4i_3}t^{a_4a_2}_{i_2i_1}\,\hat{E}^{a_1i_1i_2}_{i_3a_2i_4} \tag{90}$$

$$+\,f^{a_2}_{p_1}t^{a_2a_3}_{i_2i_1}t^{a_3a_1}_{i_4i_3}\,\hat{E}^{a_1i_1i_2}_{i_3i_4p_1} \tag{91}$$

$$-\,f^{i_4}_{p_1}t^{a_3a_1}_{i_3i_2}t^{a_3a_2}_{i_4i_1}\,\hat{E}^{a_1i_1p_1}_{i_2a_2i_3} \tag{92}$$

$$-\,f^{i_4}_{p_1}t^{a_3a_1}_{i_3i_2}t^{a_3a_2}_{i_1i_4}\,\hat{E}^{a_1i_1p_1}_{i_2i_3a_2} \tag{93}$$

each three-body term is decomposed with the grouped identity. Every template index appears in exactly one factor of each piece, so a whole operator group survives or dies together: a group survives iff its operator carries every virtual index of the pattern; in particular the scalar part (pure density matrices) always vanishes. The surviving groups are: 9 of the 19 groups survive (6 two-body, 3 one-body):

$$f^{a_2}_{p_1}t^{a_2a_3}_{i_2i_1}t^{a_3a_1}_{i_4i_3}\,\hat{E}^{a_1i_1i_2}_{i_3i_4p_1} \approx \hat{E}^{a1i1}_{i3i4}\ \big(f^{a_2}_{p_1}t^{a_2a_3}_{i_2i_1}t^{a_3a_1}_{i_4i_3}D^{i_2}_{p_1}\big) \tag{94}$$

$$\hat{E}^{a1i1}_{i3p1}\ \Big(-\tfrac{1}{2}\,f^{a_2}_{p_1}t^{a_2a_3}_{i_2i_1}t^{a_3a_1}_{i_4i_3}D^{i_2}_{i_4}\Big) \tag{95}$$

$$\hat{E}^{a1i1}_{p1i4}\ \left(-\tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_2a_3}_{i_2i_1} t^{a_3a_1}_{i_4i_3} D^{i_2}_{i_3}\right) \tag{96}$$

$$\hat{E}^{a1i2}_{i3i4}\ \left(-\tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_2a_3}_{i_2i_1} t^{a_3a_1}_{i_4i_3} D^{i_1}_{p_1}\right) \tag{97}$$

$$\hat{E}^{a1i2}_{i3p1}\ \left(f^{a_2}_{p_1} t^{a_2a_3}_{i_2i_1} t^{a_3a_1}_{i_4i_3} D^{i_1}_{i_4}\right) \tag{98}$$

$$\hat{E}^{a1i2}_{i4p1}\ \left(-\tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_2a_3}_{i_2i_1} t^{a_3a_1}_{i_4i_3} D^{i_1}_{i_3}\right) \tag{99}$$

$$\hat{E}^{a1}_{i3}\ \Big(f^{a_2}_{p_1} t^{a_2a_3}_{i_2i_1} t^{a_3a_1}_{i_4i_3} D^{i_1i_2}_{i_4p_1} \tag{100}$$

$$-2\, f^{a_2}_{p_1} t^{a_2a_3}_{i_2i_1} t^{a_3a_1}_{i_4i_3} D^{i_1}_{i_4} D^{i_2}_{p_1} \tag{101}$$

$$+f^{a_2}_{p_1} t^{a_2a_3}_{i_2i_1} t^{a_3a_1}_{i_4i_3} D^{i_1}_{p_1} D^{i_2}_{i_4}\Big) \tag{102}$$

$$\hat{E}^{a1}_{i4}\ \Big(-\tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_2a_3}_{i_2i_1} t^{a_3a_1}_{i_4i_3} D^{i_1i_2}_{i_3p_1} \tag{103}$$

$$+f^{a_2}_{p_1} t^{a_2a_3}_{i_2i_1} t^{a_3a_1}_{i_4i_3} D^{i_1}_{i_3} D^{i_2}_{p_1} \tag{104}$$

$$-\tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_2a_3}_{i_2i_1} t^{a_3a_1}_{i_4i_3} D^{i_1}_{p_1} D^{i_2}_{i_3}\Big) \tag{105}$$

$$\hat{E}^{a1}_{p1}\ \Big(-\tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_2a_3}_{i_2i_1} t^{a_3a_1}_{i_4i_3} D^{i_1}_{i_3} D^{i_2}_{i_4} \tag{106}$$

$$+f^{a_2}_{p_1} t^{a_2a_3}_{i_2i_1} t^{a_3a_1}_{i_4i_3} D^{i_1}_{i_4} D^{i_2}_{i_3} \tag{107}$$

$$-\tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_2a_3}_{i_2i_1} t^{a_3a_1}_{i_4i_3} D^{i_2i_1}_{i_3i_4}\Big) \tag{108}$$

5 of the 19 groups survive (4 two-body, 1 one-body):

$$f^{a_3}_{a_4} t^{a_3a_1}_{i_4i_3} t^{a_4a_2}_{i_2i_1} \hat{E}^{a_1i_1i_2}_{i_3a_2i_4} \approx \hat{E}^{a1i1}_{i3a2}\ \left(f^{a_3}_{a_4} t^{a_3a_1}_{i_4i_3} t^{a_4a_2}_{i_2i_1} D^{i_2}_{i_4}\right) \tag{109}$$

$$\hat{E}^{a1i1}_{i4a2}\ \left(-\tfrac{1}{2}\, f^{a_3}_{a_4} t^{a_3a_1}_{i_4i_3} t^{a_4a_2}_{i_2i_1} D^{i_2}_{i_3}\right) \tag{110}$$

$$\hat{E}^{a1i2}_{i3a2}\ \left(-\tfrac{1}{2}\, f^{a_3}_{a_4} t^{a_3a_1}_{i_4i_3} t^{a_4a_2}_{i_2i_1} D^{i_1}_{i_4}\right) \tag{111}$$

$$\hat{E}^{a1i2}_{a2i4}\ \left(-\tfrac{1}{2}\, f^{a_3}_{a_4} t^{a_3a_1}_{i_4i_3} t^{a_4a_2}_{i_2i_1} D^{i_1}_{i_3}\right) \tag{112}$$

$$\hat{E}^{a1}_{a2}\ \Big(-\tfrac{1}{2}\, f^{a_3}_{a_4} t^{a_3a_1}_{i_4i_3} t^{a_4a_2}_{i_2i_1} D^{i_1i_2}_{i_3i_4} \tag{113}$$

$$+f^{a_3}_{a_4} t^{a_3a_1}_{i_4i_3} t^{a_4a_2}_{i_2i_1} D^{i_1}_{i_3} D^{i_2}_{i_4} \tag{114}$$

$$-\tfrac{1}{2}\, f^{a_3}_{a_4} t^{a_3a_1}_{i_4i_3} t^{a_4a_2}_{i_2i_1} D^{i_1}_{i_4} D^{i_2}_{i_3}\Big) \tag{115}$$

5 of the 19 groups survive (4 two-body, 1 one-body):

$$-f^{i_4}_{p_1} t^{a_3a_1}_{i_3i_2} t^{a_3a_2}_{i_4i_1} \hat{E}^{a_1i_1p_1}_{i_2a_2i_3} \approx \hat{E}^{a1i1}_{i2a2}\ \left(-f^{i_4}_{p_1} t^{a_3a_1}_{i_3i_2} t^{a_3a_2}_{i_4i_1} D^{p_1}_{i_3}\right) \tag{116}$$

$$\hat{E}^{a1i1}_{i3a2}\ \left(\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3a_1}_{i_3i_2} t^{a_3a_2}_{i_4i_1} D^{p_1}_{i_2}\right) \tag{117}$$

$$\hat{E}^{a1p1}_{i2a2}\ \left(\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3a_1}_{i_3i_2} t^{a_3a_2}_{i_4i_1} D^{i_1}_{i_3}\right) \tag{118}$$

$$\hat{E}^{a1p1}_{a2i3}\ \left(\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3a_1}_{i_3i_2} t^{a_3a_2}_{i_4i_1} D^{i_1}_{i_2}\right) \tag{119}$$

$$\hat{E}^{a1}_{a2}\ \Big(\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3a_1}_{i_3i_2} t^{a_3a_2}_{i_4i_1} D^{i_1p_1}_{i_2i_3} \tag{120}$$

$$-f^{i_4}_{p_1} t^{a_3a_1}_{i_3i_2} t^{a_3a_2}_{i_4i_1} D^{i_1}_{i_2} D^{p_1}_{i_3} \tag{121}$$

$$+\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3a_1}_{i_3i_2} t^{a_3a_2}_{i_4i_1} D^{i_1}_{i_3} D^{p_1}_{i_2}\Big) \tag{122}$$

5 of the 19 groups survive (4 two-body, 1 one-body):

$$-f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1 p_1}_{i_2 i_3 a_2} \approx \hat{E}^{a1i1}_{i2a2} \left(\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} D^{p_1}_{i_3}\right) \tag{123}$$

$$\hat{E}^{a1i1}_{a2i3} \left(\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} D^{p_1}_{i_2}\right) \tag{124}$$

$$\hat{E}^{a1p1}_{i2a2} \left(-f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} D^{i_1}_{i_3}\right) \tag{125}$$

$$\hat{E}^{a1p1}_{i3a2} \left(\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} D^{i_1}_{i_2}\right) \tag{126}$$

$$\hat{E}^{a1}_{a2} \left(\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} D^{i_1}_{i_2} D^{p_1}_{i_3}\right. \tag{127}$$

$$-f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} D^{i_1}_{i_3} D^{p_1}_{i_2} \tag{128}$$

$$\left.+\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} D^{p_1 i_1}_{i_2 i_3}\right) \tag{129}$$

the decomposition of the three-body part produces 36 terms; the final block (surviving terms and decomposition merged) has 38 terms:

$$\left[\left[\hat{F},\, \hat{T}_2\right]^\dagger,\, \hat{T}_2\right] = -\tfrac{1}{2}\, D^{i_1 i_2}_{i_3 i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_1 i_2} t^{a_4 a_2}_{i_3 i_4} \hat{E}^{a_1}_{a_2} \tag{130}$$

$$+\tfrac{1}{2}\, D^{i_1 i_2}_{i_3 p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_2} t^{a_3 a_2}_{i_3 i_4} \hat{E}^{a_1}_{a_2} \tag{131}$$

$$+\tfrac{1}{2}\, D^{i_1 i_2}_{i_3 p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_1} t^{a_3 a_2}_{i_4 i_3} \hat{E}^{a_1}_{a_2} \tag{132}$$

$$+ D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_1 i_3} t^{a_4 a_2}_{i_2 i_4} \hat{E}^{a_1}_{a_2} \tag{133}$$

$$-\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_1 i_3} t^{a_4 a_2}_{i_4 i_2} \hat{E}^{a_1}_{a_2} \tag{134}$$

$$- D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_3} t^{a_3 a_2}_{i_2 i_4} \hat{E}^{a_1}_{a_2} \tag{135}$$

$$+\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_3} t^{a_3 a_2}_{i_4 i_2} \hat{E}^{a_1}_{a_2} \tag{136}$$

$$+\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_1} t^{a_3 a_2}_{i_2 i_4} \hat{E}^{a_1}_{a_2} \tag{137}$$

$$- D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_1} t^{a_3 a_2}_{i_4 i_2} \hat{E}^{a_1}_{a_2} \tag{138}$$

$$-\tfrac{1}{2}\, D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_2} t^{a_3 a_1}_{i_1 i_4} \hat{E}^{a_1}_{i_1} \tag{139}$$

$$+ D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_2} t^{a_3 a_1}_{i_4 i_1} \hat{E}^{a_1}_{i_1} \tag{140}$$

$$-\tfrac{1}{2}\, D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_4} t^{a_3 a_1}_{i_1 i_3} \hat{E}^{a_1}_{i_1} \tag{141}$$

$$+ D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_4} t^{a_3 a_1}_{i_3 i_1} \hat{E}^{a_1}_{i_1} \tag{142}$$

$$+ D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_2} t^{a_3 a_1}_{i_1 i_3} \hat{E}^{a_1}_{i_1} \tag{143}$$

$$- 2\, D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_2} t^{a_3 a_1}_{i_3 i_1} \hat{E}^{a_1}_{i_1} \tag{144}$$

$$-\tfrac{1}{2}\, D^{i_1 i_2}_{i_3 i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} \hat{E}^{a_1}_{p_1} \tag{145}$$

$$-\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_2 i_4} \hat{E}^{a_1}_{p_1} \tag{146}$$

$$+ D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_4 i_2} \hat{E}^{a_1}_{p_1} \tag{147}$$

$$-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_2 i_3} t^{a_4 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1}_{a_2 i_2} \tag{148}$$

$$+\tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_3} t^{a_3 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1}_{a_2 i_2} \tag{149}$$

$$-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_2 i_3} t^{a_4 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{150}$$

$$-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_3 i_2} t^{a_4 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{151}$$

$$+ D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_3 i_2} t^{a_4 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{152}$$

$$+\tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_3} t^{a_3 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{153}$$
$$+\tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{154}$$
$$-\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{155}$$
$$-\tfrac{1}{2}\, D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_4} t^{a_3 a_1}_{i_3 i_2} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{156}$$
$$+\, D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_1} t^{a_3 a_1}_{i_3 i_2} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{157}$$
$$-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_2 i_4} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{158}$$
$$+\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_4 i_2} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{159}$$
$$-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_1} t^{a_3 a_1}_{i_4 i_2} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{160}$$
$$-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_1} t^{a_3 a_1}_{i_2 i_4} \hat{E}^{a_1 i_1}_{p_1 i_2} \tag{161}$$
$$+\tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_2} t^{a_3 a_2}_{i_4 i_3} \hat{E}^{a_1 p_1}_{a_2 i_1} \tag{162}$$
$$+\tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_2} t^{a_3 a_2}_{i_3 i_4} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{163}$$
$$-\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_1} t^{a_3 a_2}_{i_3 i_4} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{164}$$
$$+\tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_1} t^{a_3 a_2}_{i_4 i_3} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{165}$$
$$+\, f^{a_1}_{a_2} t^{a_1 a_3}_{i_1 i_2} t^{a_2 a_3}_{i_3 i_4} \hat{E}^{i_1 i_2}_{i_3 i_4} \tag{166}$$
$$-\, f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_4} t^{a_1 a_2}_{i_2 i_3} \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{167}$$

grouped by the operator, so that all coefficient pieces multiplying the same $\hat{E}$ can be substituted together (12 operator groups):

$$\left[\left[\hat{F},\, \hat{T}_2\right]^{\dagger},\, \hat{T}_2\right] = \hat{E}^{a1}_{a2} \left(-\tfrac{1}{2}\, D^{i_1 i_2}_{i_3 i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_1 i_2} t^{a_4 a_2}_{i_3 i_4} \right. \tag{168}$$
$$+\tfrac{1}{2}\, D^{i_1 i_2}_{i_3 p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_2} t^{a_3 a_2}_{i_3 i_4} \tag{169}$$
$$+\tfrac{1}{2}\, D^{i_1 i_2}_{i_3 p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_1} t^{a_3 a_2}_{i_4 i_3} \tag{170}$$
$$+D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_1 i_3} t^{a_4 a_2}_{i_2 i_4} \tag{171}$$
$$-\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_1 i_3} t^{a_4 a_2}_{i_4 i_2} \tag{172}$$
$$-D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_3} t^{a_3 a_2}_{i_2 i_4} \tag{173}$$
$$+\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_3} t^{a_3 a_2}_{i_4 i_2} \tag{174}$$
$$+\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_1} t^{a_3 a_2}_{i_2 i_4} \tag{175}$$
$$\left. -D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_1} t^{a_3 a_2}_{i_4 i_2} \right) \tag{176}$$
$$\hat{E}^{a1}_{i1} \left(-\tfrac{1}{2}\, D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_2} t^{a_3 a_1}_{i_1 i_4} \right. \tag{177}$$
$$+D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_2} t^{a_3 a_1}_{i_4 i_1} \tag{178}$$
$$-\tfrac{1}{2}\, D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_4} t^{a_3 a_1}_{i_1 i_3} \tag{179}$$
$$+D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_4} t^{a_3 a_1}_{i_3 i_1} \tag{180}$$
$$+D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_2} t^{a_3 a_1}_{i_1 i_3} \tag{181}$$
$$\left. -2\, D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_2} t^{a_3 a_1}_{i_3 i_1} \right) \tag{182}$$
$$\hat{E}^{a1}_{p1} \left(-\tfrac{1}{2}\, D^{i_1 i_2}_{i_3 i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} \right. \tag{183}$$

$$-\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_2 i_4} \tag{184}$$

$$+D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_4 i_2}\Big) \tag{185}$$

$$\hat{E}^{i1a1}_{a2i2}\ \Big(-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_2 i_3} t^{a_4 a_2}_{i_4 i_1} \tag{186}$$

$$-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_3 i_2} t^{a_4 a_2}_{i_1 i_4} \tag{187}$$

$$+D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_3 i_2} t^{a_4 a_2}_{i_4 i_1} \tag{188}$$

$$+\tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_3} t^{a_3 a_2}_{i_4 i_1} \tag{189}$$

$$+\tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} \tag{190}$$

$$-D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_4 i_1}\Big) \tag{191}$$

$$\hat{E}^{i1a1}_{i2a2}\ \Big(-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_2 i_3} t^{a_4 a_2}_{i_1 i_4} \tag{192}$$

$$+\tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_3} t^{a_3 a_2}_{i_1 i_4}\Big) \tag{193}$$

$$\hat{E}^{i1a1}_{i2p1}\ \Big(-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_1} t^{a_3 a_1}_{i_2 i_4}\Big) \tag{194}$$

$$\hat{E}^{i1a1}_{i3i2}\ \Big(-\tfrac{1}{2}\, D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_4} t^{a_3 a_1}_{i_3 i_2} \tag{195}$$

$$+D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_1} t^{a_3 a_1}_{i_3 i_2}\Big) \tag{196}$$

$$\hat{E}^{i1a1}_{p1i2}\ \Big(-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_2 i_4} \tag{197}$$

$$+D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_4 i_2} \tag{198}$$

$$-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_1} t^{a_3 a_1}_{i_4 i_2}\Big) \tag{199}$$

$$\hat{E}^{i1i2}_{i3i4}\ \Big(f^{a_1}_{a_2} t^{a_1 a_3}_{i_1 i_2} t^{a_2 a_3}_{i_3 i_4}\Big) \tag{200}$$

$$\hat{E}^{p1a1}_{a2i1}\ \Big(\tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_2} t^{a_3 a_2}_{i_3 i_4} \tag{201}$$

$$-D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_1} t^{a_3 a_2}_{i_3 i_4} \tag{202}$$

$$+\tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_1} t^{a_3 a_2}_{i_4 i_3}\Big) \tag{203}$$

$$\hat{E}^{p1a1}_{i1a2}\ \Big(\tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_2} t^{a_3 a_2}_{i_4 i_3}\Big) \tag{204}$$

$$\hat{E}^{p1i1}_{i3i2}\ \Big(-f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_4} t^{a_1 a_2}_{i_2 i_3}\Big) \tag{205}$$

2 All final blocks

$\left[\hat{h},\, \hat{T}_1\right]$ **(2 terms)**

$$\left[\hat{h},\, \hat{T}_1\right] = -h^{i_1}_{p_1} t^{a_1}_{i_1} \hat{E}^{a_1}_{p_1} + h^{a_1}_{p_1} t^{a_1}_{i_1} \hat{E}^{p_1}_{i_1} \tag{206}$$

$\left[\hat{h},\, \hat{T}_2\right]$ **(1 terms)**

$$\left[\hat{h},\, \hat{T}_2\right] = h^{a_2}_{p_1} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{207}$$

$\Big[\hat{V},\,\hat{T}_1\Big]$ **(2 terms)**

$$\Big[\hat{V},\,\hat{T}_1\Big] = -g^{i_1p_1}_{p_2p_3}\,t^{a_1}_{i_1}\,\hat{E}^{a_1p_1}_{p_2p_3} \tag{208}$$
$$+\,g^{a_1p_3}_{p_1p_2}\,t^{a_1}_{i_1}\,\hat{E}^{p_1p_2}_{i_1p_3} \tag{209}$$

$\Big[\hat{V},\,\hat{T}_2\Big]$ **(16 terms)**

$$\Big[\hat{V},\,\hat{T}_2\Big] = -\tfrac{1}{2}\,D^{i_2p_1}_{p_2p_3}\,g^{a_2p_1}_{p_2p_3}\,t^{a_2a_1}_{i_1i_2}\,\hat{E}^{a_1}_{i_1} \tag{210}$$
$$+\,D^{i_2p_1}_{p_2p_3}\,g^{a_2p_1}_{p_2p_3}\,t^{a_2a_1}_{i_2i_1}\,\hat{E}^{a_1}_{i_1} \tag{211}$$
$$+\,D^{i_2}_{p_1}\,D^{p_2}_{p_3}\,g^{a_2p_2}_{p_1p_3}\,t^{a_2a_1}_{i_1i_2}\,\hat{E}^{a_1}_{i_1} \tag{212}$$
$$-\,2\,D^{i_2}_{p_1}\,D^{p_2}_{p_3}\,g^{a_2p_2}_{p_1p_3}\,t^{a_2a_1}_{i_2i_1}\,\hat{E}^{a_1}_{i_1} \tag{213}$$
$$-\,\tfrac{1}{2}\,D^{i_2}_{p_1}\,D^{p_2}_{p_3}\,g^{a_2p_2}_{p_3p_1}\,t^{a_2a_1}_{i_1i_2}\,\hat{E}^{a_1}_{i_1} \tag{214}$$
$$+\,D^{i_2}_{p_1}\,D^{p_2}_{p_3}\,g^{a_2p_2}_{p_3p_1}\,t^{a_2a_1}_{i_2i_1}\,\hat{E}^{a_1}_{i_1} \tag{215}$$
$$-\,\tfrac{1}{2}\,D^{i_1i_2}_{p_2p_3}\,g^{a_2p_1}_{p_2p_3}\,t^{a_2a_1}_{i_1i_2}\,\hat{E}^{a_1}_{p_1} \tag{216}$$
$$+\,D^{i_1}_{p_2}\,D^{i_2}_{p_3}\,g^{a_2p_1}_{p_2p_3}\,t^{a_2a_1}_{i_1i_2}\,\hat{E}^{a_1}_{p_1} \tag{217}$$
$$-\,\tfrac{1}{2}\,D^{i_1}_{p_2}\,D^{i_2}_{p_3}\,g^{a_2p_1}_{p_2p_3}\,t^{a_2a_1}_{i_2i_1}\,\hat{E}^{a_1}_{p_1} \tag{218}$$
$$+\,D^{p_2}_{p_3}\,g^{a_2p_2}_{p_1p_3}\,t^{a_2a_1}_{i_2i_1}\,\hat{E}^{a_1p_1}_{i_1i_2} \tag{219}$$
$$-\,\tfrac{1}{2}\,D^{p_2}_{p_3}\,g^{a_2p_2}_{p_3p_1}\,t^{a_2a_1}_{i_2i_1}\,\hat{E}^{a_1p_1}_{i_1i_2} \tag{220}$$
$$-\,\tfrac{1}{2}\,D^{i_2}_{p_3}\,g^{a_2p_2}_{p_1p_3}\,t^{a_2a_1}_{i_2i_1}\,\hat{E}^{a_1p_1}_{i_1p_2} \tag{221}$$
$$-\,\tfrac{1}{2}\,D^{i_2}_{p_3}\,g^{a_2p_2}_{p_3p_1}\,t^{a_2a_1}_{i_1i_2}\,\hat{E}^{a_1p_1}_{i_1p_2} \tag{222}$$
$$+\,D^{i_2}_{p_3}\,g^{a_2p_2}_{p_3p_1}\,t^{a_2a_1}_{i_2i_1}\,\hat{E}^{a_1p_1}_{i_1p_2} \tag{223}$$
$$-\,\tfrac{1}{2}\,D^{i_2}_{p_3}\,g^{a_2p_2}_{p_1p_3}\,t^{a_2a_1}_{i_1i_2}\,\hat{E}^{a_1p_1}_{p_2i_1} \tag{224}$$
$$+\,\tfrac{1}{2}\,g^{a_1a_2}_{p_1p_2}\,t^{a_1a_2}_{i_1i_2}\,\hat{E}^{p_1p_2}_{i_1i_2} \tag{225}$$

$\Big[\Big[\hat{F},\,\hat{T}_1\Big],\,\hat{T}_1\Big]$ **(1 terms)**

$$\Big[\Big[\hat{F},\,\hat{T}_1\Big],\,\hat{T}_1\Big] = -2\,f^{a_2}_{i_2}\,t^{a_1}_{i_2}\,t^{a_2}_{i_1}\,\hat{E}^{a_1}_{i_1} \tag{226}$$

$\Big[\Big[\hat{F},\,\hat{T}_1\Big],\,\hat{T}_2\Big]$ **(1 terms)**

$$\Big[\Big[\hat{F},\,\hat{T}_1\Big],\,\hat{T}_2\Big] = -f^{a_3}_{i_3}\,t^{a_1}_{i_3}\,t^{a_3a_2}_{i_1i_2}\,\hat{E}^{a_1a_2}_{i_1i_2} \tag{227}$$

$\Big[\Big[\hat{F},\,\hat{T}_2\Big],\,\hat{T}_1\Big]$ **(1 terms)**

$$\Big[\Big[\hat{F},\,\hat{T}_2\Big],\,\hat{T}_1\Big] = -f^{a_3}_{i_3}\,t^{a_1}_{i_3}\,t^{a_3a_2}_{i_1i_2}\,\hat{E}^{a_1a_2}_{i_1i_2} \tag{228}$$

$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right]$ **(0 terms)**

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right] = 0 \tag{229}$$

$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_1\right]$ **(4 terms)**

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_1\right] = f^{i_1}_{i_2} t^{a_1}_{i_1} t^{a_2}_{i_2} \hat{E}^{a_1}_{a_2} \tag{230}$$
$$- f^{a_2}_{p_1} t^{a_1}_{i_1} t^{a_2}_{i_1} \hat{E}^{a_1}_{p_1} + f^{a_1}_{a_2} t^{a_1}_{i_1} t^{a_2}_{i_2} \hat{E}^{i_1}_{i_2} \tag{231}$$
$$- f^{i_2}_{p_1} t^{a_1}_{i_1} t^{a_1}_{i_2} \hat{E}^{p_1}_{i_1} \tag{232}$$

$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right]$ **(2 terms)**

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right] = f^{a_2}_{a_3} t^{a_2 a_1}_{i_3 i_2} t^{a_3}_{i_1} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{233}$$
$$- f^{i_3}_{p_1} t^{a_2 a_1}_{i_2 i_1} t^{a_2}_{i_3} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{234}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right]$ **(6 terms)**

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right] = -f^{a_3}_{p_1} t^{a_1}_{i_2} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1 i_1}_{a_2 p_1} \tag{235}$$
$$- f^{a_3}_{p_1} t^{a_1}_{i_2} t^{a_3 a_2}_{i_2 i_1} \hat{E}^{a_1 i_1}_{p_1 a_2} \tag{236}$$
$$+ f^{a_2}_{a_3} t^{a_2 a_1}_{i_2 i_1} t^{a_3}_{i_3} \hat{E}^{i_1 i_2}_{a_1 i_3} \tag{237}$$
$$+ f^{a_1}_{p_1} t^{a_1 a_2}_{i_2 i_1} t^{a_2}_{i_3} \hat{E}^{i_1 i_2}_{i_3 p_1} \tag{238}$$
$$- f^{i_3}_{p_1} t^{a_2 a_1}_{i_3 i_1} t^{a_2}_{i_2} \hat{E}^{i_1 p_1}_{a_1 i_2} \tag{239}$$
$$- f^{i_3}_{p_1} t^{a_2 a_1}_{i_1 i_3} t^{a_2}_{i_2} \hat{E}^{i_1 p_1}_{i_2 a_1} \tag{240}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right]$ **(38 terms)**

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right] = -\tfrac{1}{2}\, D^{i_1 i_2}_{i_3 i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_1 i_2} t^{a_4 a_2}_{i_3 i_4} \hat{E}^{a_1}_{a_2} \tag{241}$$
$$+ \tfrac{1}{2}\, D^{i_1 i_2}_{i_3 p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_2} t^{a_3 a_2}_{i_3 i_4} \hat{E}^{a_1}_{a_2} \tag{242}$$
$$+ \tfrac{1}{2}\, D^{i_1 i_2}_{i_3 p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_1} t^{a_3 a_2}_{i_4 i_3} \hat{E}^{a_1}_{a_2} \tag{243}$$
$$+ D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_1 i_3} t^{a_4 a_2}_{i_2 i_4} \hat{E}^{a_1}_{a_2} \tag{244}$$
$$- \tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_1 i_3} t^{a_4 a_2}_{i_4 i_2} \hat{E}^{a_1}_{a_2} \tag{245}$$

$$- D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_3} t^{a_3 a_2}_{i_2 i_4} \hat{E}^{a_1}_{a_2} \tag{246}$$

$$+ \tfrac{1}{2} D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_3} t^{a_3 a_2}_{i_4 i_2} \hat{E}^{a_1}_{a_2} \tag{247}$$

$$+ \tfrac{1}{2} D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_1} t^{a_3 a_2}_{i_2 i_4} \hat{E}^{a_1}_{a_2} \tag{248}$$

$$- D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_1} t^{a_3 a_2}_{i_4 i_2} \hat{E}^{a_1}_{a_2} \tag{249}$$

$$- \tfrac{1}{2} D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_2} t^{a_3 a_1}_{i_1 i_4} \hat{E}^{a_1}_{i_1} \tag{250}$$

$$+ D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_2} t^{a_3 a_1}_{i_4 i_1} \hat{E}^{a_1}_{i_1} \tag{251}$$

$$- \tfrac{1}{2} D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_4} t^{a_3 a_1}_{i_1 i_3} \hat{E}^{a_1}_{i_1} \tag{252}$$

$$+ D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_4} t^{a_3 a_1}_{i_3 i_1} \hat{E}^{a_1}_{i_1} \tag{253}$$

$$+ D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_2} t^{a_3 a_1}_{i_1 i_3} \hat{E}^{a_1}_{i_1} \tag{254}$$

$$- 2 D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_2} t^{a_3 a_1}_{i_3 i_1} \hat{E}^{a_1}_{i_1} \tag{255}$$

$$- \tfrac{1}{2} D^{i_1 i_2}_{i_3 i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} \hat{E}^{a_1}_{p_1} \tag{256}$$

$$- \tfrac{1}{2} D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_2 i_4} \hat{E}^{a_1}_{p_1} \tag{257}$$

$$+ D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_4 i_2} \hat{E}^{a_1}_{p_1} \tag{258}$$

$$- \tfrac{1}{2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_2 i_3} t^{a_4 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1}_{a_2 i_2} \tag{259}$$

$$+ \tfrac{1}{2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_3} t^{a_3 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1}_{a_2 i_2} \tag{260}$$

$$- \tfrac{1}{2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_2 i_3} t^{a_4 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{261}$$

$$- \tfrac{1}{2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_3 i_2} t^{a_4 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{262}$$

$$+ D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_3 i_2} t^{a_4 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{263}$$

$$+ \tfrac{1}{2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_3} t^{a_3 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{264}$$

$$+ \tfrac{1}{2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{265}$$

$$- D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{266}$$

$$- \tfrac{1}{2} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_4} t^{a_3 a_1}_{i_3 i_2} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{267}$$

$$+ D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_1} t^{a_3 a_1}_{i_3 i_2} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{268}$$

$$- \tfrac{1}{2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_2 i_4} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{269}$$

$$+ D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_4 i_2} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{270}$$

$$- \tfrac{1}{2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_1} t^{a_3 a_1}_{i_4 i_2} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{271}$$

$$- \tfrac{1}{2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_1} t^{a_3 a_1}_{i_2 i_4} \hat{E}^{a_1 i_1}_{p_1 i_2} \tag{272}$$

$$+ \tfrac{1}{2} D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_2} t^{a_3 a_2}_{i_4 i_3} \hat{E}^{a_1 p_1}_{a_2 i_1} \tag{273}$$

$$+ \tfrac{1}{2} D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_2} t^{a_3 a_2}_{i_3 i_4} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{274}$$

$$- D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_1} t^{a_3 a_2}_{i_3 i_4} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{275}$$

$$+ \tfrac{1}{2} D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_1} t^{a_3 a_2}_{i_4 i_3} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{276}$$

$$+ f^{a_1}_{a_2} t^{a_1 a_3}_{i_1 i_2} t^{a_2 a_3}_{i_3 i_4} \hat{E}^{i_1 i_2}_{i_3 i_4} \tag{277}$$

$$- f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_4} t^{a_1 a_2}_{i_2 i_3} \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{278}$$

3 All final blocks, grouped by operator

Every block with the coefficient pieces collected per operator, ready for substitution.

$\left[\hat{h},\, \hat{T}_1\right]$ **(2 terms, 2 operator groups)**

$$\left[\hat{h},\, \hat{T}_1\right] = \hat{E}^{a1}_{p1}\ \left(-h^{i_1}_{p_1} t^{a_1}_{i_1}\right) \tag{279}$$

$$\hat{E}^{p1}_{i1}\ \left(h^{a_1}_{p_1} t^{a_1}_{i_1}\right) \tag{280}$$

$\left[\hat{h},\, \hat{T}_2\right]$ **(1 terms, 1 operator groups)**

$$\left[\hat{h},\, \hat{T}_2\right] = \hat{E}^{p1a1}_{i2i1}\ \left(h^{a_2}_{p_1} t^{a_2a_1}_{i_2i_1}\right) \tag{281}$$

$\left[\hat{V},\, \hat{T}_1\right]$ **(2 terms, 2 operator groups)**

$$\left[\hat{V},\, \hat{T}_1\right] = \hat{E}^{p1a1}_{p3p2}\ \left(-g^{i_1p_1}_{p_2p_3} t^{a_1}_{i_1}\right) \tag{282}$$

$$\hat{E}^{p2p1}_{p3i1}\ \left(g^{a_1p_3}_{p_1p_2} t^{a_1}_{i_1}\right) \tag{283}$$

$\left[\hat{V},\, \hat{T}_2\right]$ **(16 terms, 6 operator groups)**

$$\left[\hat{V},\, \hat{T}_2\right] = \hat{E}^{a1}_{i1}\ \left(-\tfrac{1}{2}\, D^{i_2p_1}_{p_2p_3} g^{a_2p_1}_{p_2p_3} t^{a_2a_1}_{i_1i_2}\right. \tag{284}$$

$$+D^{i_2p_1}_{p_2p_3} g^{a_2p_1}_{p_2p_3} t^{a_2a_1}_{i_2i_1} \tag{285}$$

$$+D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_2}_{p_1p_3} t^{a_2a_1}_{i_1i_2} \tag{286}$$

$$-2\, D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_2}_{p_1p_3} t^{a_2a_1}_{i_2i_1} \tag{287}$$

$$-\tfrac{1}{2}\, D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_2}_{p_3p_1} t^{a_2a_1}_{i_1i_2} \tag{288}$$

$$\left.+D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_2}_{p_3p_1} t^{a_2a_1}_{i_2i_1}\right) \tag{289}$$

$$\hat{E}^{a1}_{p1}\ \left(-\tfrac{1}{2}\, D^{i_1i_2}_{p_2p_3} g^{a_2p_1}_{p_2p_3} t^{a_2a_1}_{i_1i_2}\right. \tag{290}$$

$$+D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2p_1}_{p_2p_3} t^{a_2a_1}_{i_1i_2} \tag{291}$$

$$\left.-\tfrac{1}{2}\, D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2p_1}_{p_2p_3} t^{a_2a_1}_{i_2i_1}\right) \tag{292}$$

$$\hat{E}^{p1a1}_{i1p2}\ \left(-\tfrac{1}{2}\, D^{i_2}_{p_3} g^{a_2p_2}_{p_1p_3} t^{a_2a_1}_{i_1i_2}\right) \tag{293}$$

$$\hat{E}^{p1a1}_{i2i1}\ \left(D^{p_2}_{p_3} g^{a_2p_2}_{p_1p_3} t^{a_2a_1}_{i_2i_1}\right. \tag{294}$$

$$\left.-\tfrac{1}{2}\, D^{p_2}_{p_3} g^{a_2p_2}_{p_3p_1} t^{a_2a_1}_{i_2i_1}\right) \tag{295}$$

$$\hat{E}^{p1a1}_{p2i1}\ \left(-\tfrac{1}{2}\, D^{i_2}_{p_3} g^{a_2p_2}_{p_1p_3} t^{a_2a_1}_{i_2i_1}\right. \tag{296}$$

$$-\tfrac{1}{2}\, D^{i_2}_{p_3} g^{a_2p_2}_{p_3p_1} t^{a_2a_1}_{i_1i_2} \tag{297}$$

$$\left.+D^{i_2}_{p_3} g^{a_2p_2}_{p_3p_1} t^{a_2a_1}_{i_2i_1}\right) \tag{298}$$

$$\hat{E}^{p1p2}_{i1i2}\ \left(\tfrac{1}{2}\, g^{a_1a_2}_{p_1p_2} t^{a_1a_2}_{i_1i_2}\right) \tag{299}$$

$\left[\left[\hat{F},\hat{T}_1\right],\hat{T}_1\right]$ **(1 terms, 1 operator groups)**

$$\left[\left[\hat{F},\hat{T}_1\right],\hat{T}_1\right] = \hat{E}^{a1}_{i1}\ \left(-2\, f^{a_2}_{i_2} t^{a_1}_{i_2} t^{a_2}_{i_1}\right) \tag{300}$$

$\left[\left[\hat{F},\hat{T}_1\right],\hat{T}_2\right]$ **(1 terms, 1 operator groups)**

$$\left[\left[\hat{F},\hat{T}_1\right],\hat{T}_2\right] = \hat{E}^{a1a2}_{i1i2}\ \left(-f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_3a_2}_{i_1i_2}\right) \tag{301}$$

$\left[\left[\hat{F},\hat{T}_2\right],\hat{T}_1\right]$ **(1 terms, 1 operator groups)**

$$\left[\left[\hat{F},\hat{T}_2\right],\hat{T}_1\right] = \hat{E}^{a1a2}_{i1i2}\ \left(-f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_3a_2}_{i_1i_2}\right) \tag{302}$$

$\left[\left[\hat{F},\hat{T}_2\right],\hat{T}_2\right]$ **(0 terms, 0 operator groups)**

$$\left[\left[\hat{F},\hat{T}_2\right],\hat{T}_2\right] = 0 \tag{303}$$

$\left[\left[\hat{F},\hat{T}_1\right]^\dagger,\hat{T}_1\right]$ **(4 terms, 4 operator groups)**

$$\left[\left[\hat{F},\hat{T}_1\right]^\dagger,\hat{T}_1\right] = \hat{E}^{a1}_{a2}\ \left(f^{i_1}_{i_2} t^{a_1}_{i_1} t^{a_2}_{i_2}\right) \tag{304}$$

$$\hat{E}^{a1}_{p1}\ \left(-f^{a_2}_{p_1} t^{a_1}_{i_1} t^{a_2}_{i_1}\right) \tag{305}$$

$$\hat{E}^{i1}_{i2}\ \left(f^{a_1}_{a_2} t^{a_1}_{i_1} t^{a_2}_{i_2}\right) \tag{306}$$

$$\hat{E}^{p1}_{i1}\ \left(-f^{i_2}_{p_1} t^{a_1}_{i_1} t^{a_1}_{i_2}\right) \tag{307}$$

$\left[\left[\hat{F},\hat{T}_1\right]^\dagger,\hat{T}_2\right]$ **(2 terms, 2 operator groups)**

$$\left[\left[\hat{F},\hat{T}_1\right]^\dagger,\hat{T}_2\right] = \hat{E}^{i1a1}_{i3i2}\ \left(f^{a_2}_{a_3} t^{a_2a_1}_{i_3i_2} t^{a_3}_{i_1}\right) \tag{308}$$

$$\hat{E}^{p1a1}_{i2i1}\ \left(-f^{i_3}_{p_1} t^{a_2a_1}_{i_2i_1} t^{a_2}_{i_3}\right) \tag{309}$$

$\left[\left[\hat{F},\hat{T}_2\right]^\dagger,\hat{T}_1\right]$ **(6 terms, 6 operator groups)**

$$\left[\left[\hat{F},\hat{T}_2\right]^\dagger,\hat{T}_1\right] = \hat{E}^{i1a1}_{a2p1}\ \left(-f^{a_3}_{p_1} t^{a_1}_{i_2} t^{a_3a_2}_{i_2i_1}\right) \tag{310}$$

$$\hat{E}^{i1a1}_{p1a2}\ \left(-f^{a_3}_{p_1} t^{a_1}_{i_2} t^{a_3a_2}_{i_1i_2}\right) \tag{311}$$

$$\hat{E}^{i2i1}_{i3a1}\ \left(f^{a_2}_{a_3} t^{a_2 a_1}_{i_2 i_1} t^{a_3}_{i_3}\right) \tag{312}$$
$$\hat{E}^{i2i1}_{p1i3}\ \left(f^{a_1}_{p_1} t^{a_1 a_2}_{i_2 i_1} t^{a_2}_{i_3}\right) \tag{313}$$
$$\hat{E}^{p1i1}_{a1i2}\ \left(-f^{i_3}_{p_1} t^{a_2 a_1}_{i_1 i_3} t^{a_2}_{i_2}\right) \tag{314}$$
$$\hat{E}^{p1i1}_{i2a1}\ \left(-f^{i_3}_{p_1} t^{a_2 a_1}_{i_3 i_1} t^{a_2}_{i_2}\right) \tag{315}$$

$\left[\left[\hat{F},\, \hat{T}_2\right]^\dagger,\, \hat{T}_2\right]$ **(38 terms, 12 operator groups)**

$$\left[\left[\hat{F},\, \hat{T}_2\right]^\dagger,\, \hat{T}_2\right] = \hat{E}^{a1}_{a2}\ \Big(-\tfrac{1}{2}\, D^{i_1 i_2}_{i_3 i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_1 i_2} t^{a_4 a_2}_{i_3 i_4} \tag{316}$$
$$+\tfrac{1}{2}\, D^{i_1 i_2}_{i_3 p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_2} t^{a_3 a_2}_{i_3 i_4} \tag{317}$$
$$+\tfrac{1}{2}\, D^{i_1 i_2}_{i_3 p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_1} t^{a_3 a_2}_{i_4 i_3} \tag{318}$$
$$+D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_1 i_3} t^{a_4 a_2}_{i_2 i_4} \tag{319}$$
$$-\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_1 i_3} t^{a_4 a_2}_{i_4 i_2} \tag{320}$$
$$-D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_3} t^{a_3 a_2}_{i_2 i_4} \tag{321}$$
$$+\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_3} t^{a_3 a_2}_{i_4 i_2} \tag{322}$$
$$+\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_1} t^{a_3 a_2}_{i_2 i_4} \tag{323}$$
$$-D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_1} t^{a_3 a_2}_{i_4 i_2}\Big) \tag{324}$$
$$\hat{E}^{a1}_{i1}\ \Big(-\tfrac{1}{2}\, D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_2} t^{a_3 a_1}_{i_1 i_4} \tag{325}$$
$$+D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_2} t^{a_3 a_1}_{i_4 i_1} \tag{326}$$
$$-\tfrac{1}{2}\, D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_4} t^{a_3 a_1}_{i_1 i_3} \tag{327}$$
$$+D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_4} t^{a_3 a_1}_{i_3 i_1} \tag{328}$$
$$+D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_2} t^{a_3 a_1}_{i_1 i_3} \tag{329}$$
$$-2\, D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_2} t^{a_3 a_1}_{i_3 i_1}\Big) \tag{330}$$
$$\hat{E}^{a1}_{p1}\ \Big(-\tfrac{1}{2}\, D^{i_1 i_2}_{i_3 i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} \tag{331}$$
$$-\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_2 i_4} \tag{332}$$
$$+D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_4 i_2}\Big) \tag{333}$$
$$\hat{E}^{i1a1}_{a2i2}\ \Big(-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_2 i_3} t^{a_4 a_2}_{i_4 i_1} \tag{334}$$
$$-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_3 i_2} t^{a_4 a_2}_{i_1 i_4} \tag{335}$$
$$+D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_3 i_2} t^{a_4 a_2}_{i_4 i_1} \tag{336}$$
$$+\tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_3} t^{a_3 a_2}_{i_4 i_1} \tag{337}$$
$$+\tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} \tag{338}$$
$$-D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_4 i_1}\Big) \tag{339}$$
$$\hat{E}^{i1a1}_{i2a2}\ \Big(-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_2 i_3} t^{a_4 a_2}_{i_1 i_4} \tag{340}$$
$$+\tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_3} t^{a_3 a_2}_{i_1 i_4}\Big) \tag{341}$$

$$\hat{E}^{i1a1}_{i2p1}\ \left(-\tfrac{1}{2}\,D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2a_3}_{i_3i_1} t^{a_3a_1}_{i_2i_4}\right) \tag{342}$$

$$\hat{E}^{i1a1}_{i3i2}\ \left(-\tfrac{1}{2}\,D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2a_3}_{i_1i_4} t^{a_3a_1}_{i_3i_2}\right. \tag{343}$$

$$\left.+D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2a_3}_{i_4i_1} t^{a_3a_1}_{i_3i_2}\right) \tag{344}$$

$$\hat{E}^{i1a1}_{p1i2}\ \left(-\tfrac{1}{2}\,D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2a_3}_{i_1i_3} t^{a_3a_1}_{i_2i_4}\right. \tag{345}$$

$$+D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2a_3}_{i_1i_3} t^{a_3a_1}_{i_4i_2} \tag{346}$$

$$\left.-\tfrac{1}{2}\,D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2a_3}_{i_3i_1} t^{a_3a_1}_{i_4i_2}\right) \tag{347}$$

$$\hat{E}^{i1i2}_{i3i4}\ \left(f^{a_1}_{a_2} t^{a_1a_3}_{i_1i_2} t^{a_2a_3}_{i_3i_4}\right) \tag{348}$$

$$\hat{E}^{p1a1}_{a2i1}\ \left(\tfrac{1}{2}\,D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3a_1}_{i_1i_2} t^{a_3a_2}_{i_3i_4}\right. \tag{349}$$

$$-D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3a_1}_{i_2i_1} t^{a_3a_2}_{i_3i_4} \tag{350}$$

$$\left.+\tfrac{1}{2}\,D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3a_1}_{i_2i_1} t^{a_3a_2}_{i_4i_3}\right) \tag{351}$$

$$\hat{E}^{p1a1}_{i1a2}\ \left(\tfrac{1}{2}\,D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3a_1}_{i_1i_2} t^{a_3a_2}_{i_4i_3}\right) \tag{352}$$

$$\hat{E}^{p1i1}_{i3i2}\ \left(-f^{i_4}_{p_1} t^{a_1a_2}_{i_1i_4} t^{a_1a_2}_{i_2i_3}\right) \tag{353}$$